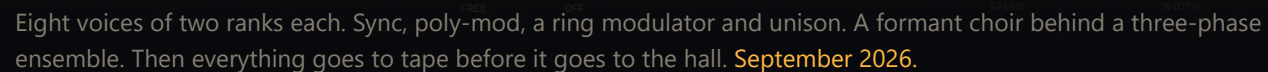


1984

BUILD 260923.1



EIGHT VOICES, SIXTEEN RANKS, ONE TAPE



The deck. Rank I above, rank II below, each a row of long-throw sliders in the five cap colours. The amber display names the patch and draws the filter envelopes and the eight voice lamps. The chain runs left to right along the bottom, and the tape reels turn when the tape is on.

INSTALLING

Copy **1984.vst3** into your VST3 folder — normally **C:\PROGRAM FILES\COMMON FILES\VST3** — and rescan. The standalone in the same download needs nothing installing.

A voice in 1984 is not one synthesizer. It is **two**, called rank I and rank II, sharing a keyboard, a ring modulator and a sub-oscillator. Each rank has its own oscillator, its own resonant high-pass into its own resonant low-pass, its own filter envelope, its own amplifier envelope, and its own level and pan. Eight voices is therefore sixteen complete ranks, and every patch on the dial is two instruments played as one.

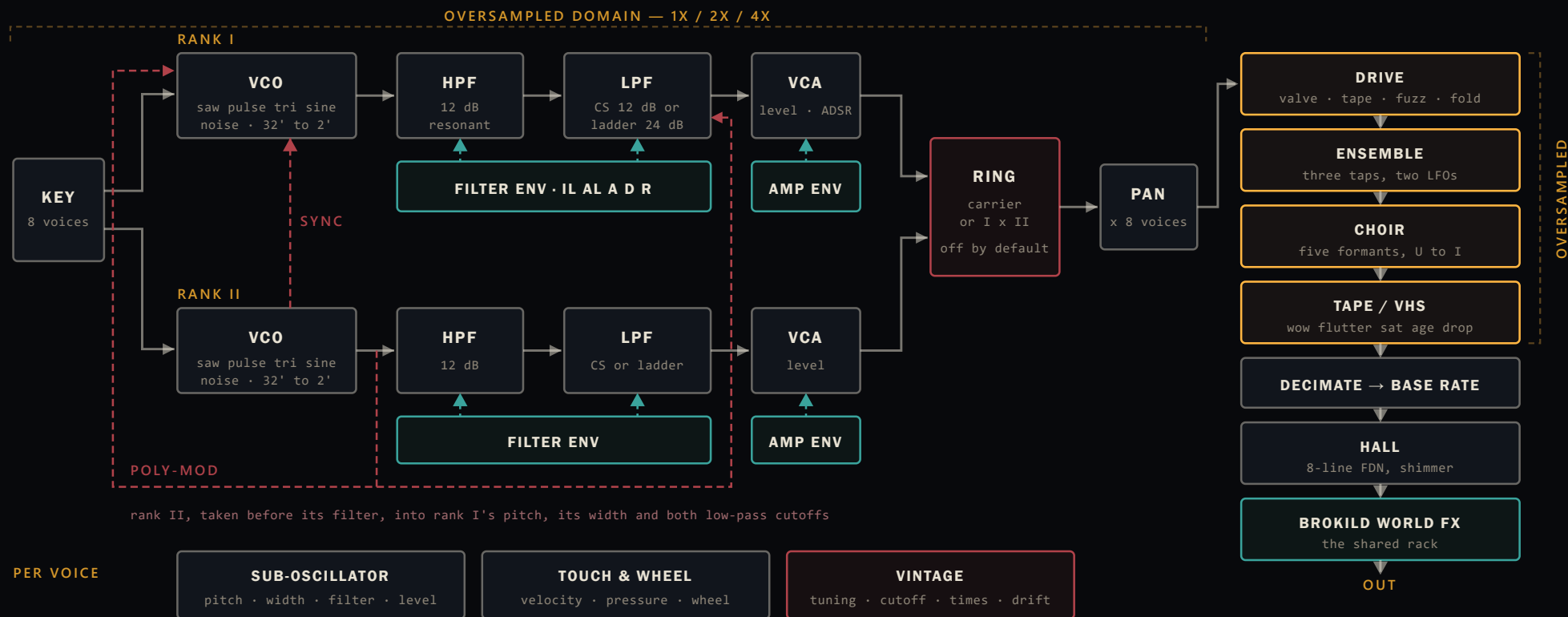
The architecture is the one the big polysynth of 1977 used and nobody has really used since: two ranks per voice, a high-pass and a low-pass in series, and a filter envelope with an **initial level** and an **attack level** instead of a sustain. The strange part — hard sync, poly-mod, unison — comes from the other side of the Atlantic. The choir is a fixed formant bank behind a three-phase ensemble.

The eight voices are summed and go through one chain: drive, ensemble, choir, tape, hall, and then the Brokild World FX rack. Everything up to and including the tape runs oversampled, so nothing that saturates can alias. The hall runs at the base rate, because it is linear and long.

THE FIRST MINUTE

Dial up **CS STRINGS** and hold a chord. Then take **UNISON DETUNE** up, put the voice mode on **DUO**, and listen to eight oscillator boards instead of four. Everything else in this handbook is detail.

THE SIGNAL PATH

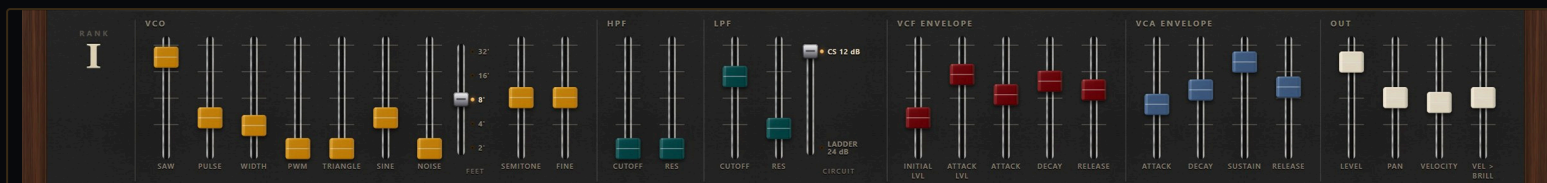


Solid lines are audio, broken red lines are modulation, dotted teal lines are envelopes. One filter envelope drives both filters of its rank, and that is the thing that makes this architecture sound the way it does: the high-pass and the low-pass move together, so the note changes shape from both ends at once.

Rank II's wrap resets rank I's phase when **SYNC** is on: the audible rank is the slave, the way the Prophet does it. Rank II's output, taken **before** its filter, can bend rank I's pitch, its pulse width and both low-pass cutoffs — so rank II can be an audible layer, a silent modulator, or both at once.

Everything up to and including the tape runs oversampled, so nothing that saturates can fold back into the band. The hall runs at the base rate, because it is linear and long and gains nothing from the extra work. The whole voice is drawn once here; there are eight of them.

THE OSCILLATOR



One rank strip, end to end. Waves and tuning, the two filters, the two envelopes, and level, pan and velocity. Rank II carries the same strip again, plus SYNC.

One phase accumulator per rank, four band-limited waves out of it at once, each with its own slider. They are not a selector: saw plus a little pulse plus a little sine is one sound, and it is the brass recipe.

SAW · PULSE · TRIANGLE · SINE

FOUR LEVELS, NOT A SWITCH

Mix them. The saw is the body, the pulse the reed, the triangle the flute, and the sine weight underneath. The sine is taken after the filter, so it survives when everything else has been shut away.

WIDTH · PWM

50 TO 95 PER CENT

WIDTH sets the pulse; **PWM** hands it to the sub-oscillator. A slow PWM on two ranks is most of a string patch.

NOISE

WHITE, PER VOICE, INTO THE FILTER

It goes in ahead of the high-pass, so the filters shape it exactly as they shape the oscillator.

FEET · SEMITONE · FINE

32' 16' 8' 4' 2' · ±12 SEMITONES · ±50 CENTS

Nearly every factory patch puts rank II a handful of cents off rank I and leaves it there: that beat is the sound of two oscillator boards.

SYNC

SYNC I TO II resets rank I's phase every time rank II wraps, band-limited at the actual jump height — so a synced rank is a hard edge and not a burst of aliasing. Rank II is the master because the envelope's pitch route lands on rank I: turn rank II's level down, and the envelope sweeps the synced, audible rank over a note the silent master holds.

MEASURED

Both ranks sit within **one cent** at every footage. A saw at A6 keeps its worst non-harmonic component below **-60 dB** at 2x.

TWO FILTERS, ONE ENVELOPE

HPF · HPF RES

10 HZ TO 12 KHZ, 12 DB PER OCTAVE

A resonant high-pass ahead of the low-pass. This is what makes a string patch thin and a choir patch hollow, and it is the half of the architecture that most synthesizers do not have. Push the resonance and it will sing; the band-pass state is softly limited so a screaming resonance stays a note instead of a spike.

LPF · LPF RES

20 HZ TO 20 KHZ

The main filter. Resonance runs to self-oscillation in both circuits, and the last few per cent of the sweep are where it crosses over into oscillating on its own.

FILTER

TWO CIRCUITS, PER RANK

CS 12 DB A state-variable pair. Open, glassy, and the one that lets the high-pass be heard.

LADDER 24 DB Four poles with a soft saturator at the summing node. Thicker, and it takes a bass line down where a 12 dB filter cannot.

The two ranks can run different circuits. A ladder rank under a CS rank is most of what the brass patches are doing.

WHAT SETS A CUTOFF

CUTOFF = `slider × 2(env + key + brilliance + velocity + touch + sub + poly-mod)`

Everything that opens a filter does it in octaves and they add. The filter envelope alone is worth up to six of them.

BRILLIANCE · RESONANCE

GLOBAL, BIPOLAR, ON BOTH RANKS

Two performance trims that ride over whatever the patch set:

BRILLIANCE shifts both cutoffs, **RESONANCE** scales both resonances.

They read from −100 to +100 per cent, with the centre meaning "as the patch has it".

KEY TRACK

HOW FAR THE CUTOFF FOLLOWS THE NOTE

At zero the filter stays put and the top of the keyboard goes dull; at full it follows the note and the timbre is even everywhere.

MEASURED

Asked to self-oscillate at 1 kHz, the state-variable filter sits at **1000.0 Hz** and the ladder at **996.5 Hz** — 0.35 per cent — and both stay bounded. Resonance at 80 per cent puts the cutoff **14.5 dB** over the octave above it.

THE ENVELOPES, AND THE END OF THE RANK



INITIAL LEVEL · ATTACK LEVEL

BOTH BIPOLAR, -100 TO +100 PER CENT

The filter envelope has no sustain, because **IL** and **AL** are the sustain. At key down it jumps to IL, moves to AL over the attack time, then decays to zero. Both are bipolar, so a filter can open from below (IL negative, AL positive) or slam shut from above.

VCF ATTACK · DECAY · RELEASE

1 MS TO 10 S · 2 MS TO 20 S

Release runs from wherever the envelope had got to when you let go, not from a fixed point — so a short note releases from high up and a held one from near zero.

VCA ATTACK · DECAY · SUSTAIN · RELEASE

THE ORDINARY ONE

An RC-shaped ADSR that retriggers from its current level, so a repeated note does not click. Sustain at 100 with a long decay is an organ; sustain at 0 with a short decay is a bell.

LEVEL · PAN · VELOCITY · VEL>BRILL

THE END OF THE STRIP

LEVEL is the rank in the voice — set rank II to zero and it is a silent modulator. **PAN** places it; the global **STEREO SPREAD** widens the eight voices around it. **VELOCITY** is how hard you have to play for full level, and **VEL>BRILL** hands velocity to the filter instead.

TOUCH, WHEEL AND THE SUB-OSCILLATOR



The modulation row. Sub-oscillator, ring modulator, poly-mod, and the touch and wheel depths.

TOUCH

AFTERTOUCH, CHANNEL OR POLYPHONIC

Lean into a held key and the note answers. Four destinations, each with its own depth: **TOUCH>PITCH**, **TOUCH>BRILL**, **TOUCH>LEVEL** and **TOUCH>SUB**, which brings in vibrato only when you press. If the keyboard sends both channel and polyphonic pressure, the higher of the two wins for that note.

WHEEL

VIBRATO, BRILLIANCE, RATE

WHEEL>VIBRATO and **WHEEL>BRILL** with a dedicated **WHEEL RATE** from 2 to 10 Hz, so the wheel's vibrato is its own speed and does not disturb whatever the sub-oscillator is already doing.

THE SUB-OSCILLATOR

ONE PER VOICE · 0.05 TO 40 HZ · SEVEN SHAPES

Sine, triangle, saw, ramp, square, sample-and-hold and smooth noise, with a delay from 10 ms to 5 s so vibrato arrives after the note has settled. Four destinations:

Sub>Pitch	vibrato
Sub>Width	PWM — the string sound
Sub>Filter	a growl, or a very slow tide
Sub>Level	tremolo

SUB PHASE — THE ONE THAT MATTERS

FREE	Every voice runs its own. Eight voices wandering independently is a section of players.
RESET	From zero at each note-on. Tight, deliberate, good for a lead.
ONE	All eight share a phase. A chord modulated in lockstep is a synthesizer on a tape.

This is one switch and it changes what the instrument **is**. Try the same pad on **FREE** and on **ONE**.

RING MODULATOR & POLY-MOD

RING — THREE MODES

OFF	Where it sits on a fresh instrument.
CARRIER	A sine carrier of its own, 0.1 Hz to 4 kHz. Slow it down and it is tremolo; speed it up and it is metal.
RANKS	Rank I multiplied by rank II. Tune them apart and you get sums and differences that have nothing to do with the key.

KEY · SPEED · DEPTH

THE CARRIER

With **KEY** off, **SPEED** is a fixed frequency in hertz and the ring is inharmonic everywhere on the keyboard — which is exactly what a bell wants. With **KEY** on, **SPEED** becomes a ratio of the note, from a quarter to eight times, and the modulation tracks the key. **DEPTH** crossfades from dry to fully ring-modulated.

ATTACK · DECAY · MOD

THE CARRIER HAS AN ENVELOPE OF ITS OWN

MOD is bipolar: positive sweeps the carrier up over the note, negative sweeps it down. A downward sweep with a long decay is a struck bar; an upward one is the sound of a machine starting.

POLY-MOD

TWO SOURCES, FIVE DESTINATIONS

Modulation inside the voice, at audio rate, per voice — so eight notes each modulate themselves rather than sharing one LFO. There are two sources. One is **rank II, taken before its filter**; the other is **rank I's own filter envelope**.

II>I Pitch	exponential FM at audio rate. A little is a growl; a lot is a klaxon.
II>I Width	rank II opens and shuts rank I's pulse, cycle by cycle.
II>Filter	filter FM into both low-passes.
Env>I Pitch	bipolar. The classic sweep: the envelope bends the note.
Env>I Width	bipolar. The pulse opens as the filter does.

SET RANK II'S LEVEL TO ZERO

Poly-mod takes rank II from **before** its amplifier, so its **LEVEL** slider decides only whether you also hear it. At zero it is a pure modulator and rank I is the whole sound. Leave it up and it is both at once. **POLY-MOD SWEEP** on the patch dial is the first case; go and look at it.

VOICE MODES, GLIDE, VINTAGE

VOICE MODE

POLY	Eight voices, one per key. A ninth note steals the oldest.
DUO	Two voices per key, detuned and panned apart: four-note polyphony, but every note is a pair of players.
UNISON	All eight on one note, fanned symmetrically around it by UNISON DETUNE, up to 60 cents.
MONO	Last note wins, with legato if you want it.

DUO and UNISON both read **UNISON DETUNE**. It is the single most effective control on the panel for making an eight-voice machine sound like a wall.

GLIDE · GLISSANDO · LEGATO

5 MS TO 5 S, OR OFF

GLISSANDO makes the slide step through semitones instead of sweeping. **LEGATO** only glides when you are already holding a note, so a phrase slides and a fresh attack does not.

VINTAGE

ONE KNOB, FIVE TOLERANCES

At zero the eight voices are identical, which no analogue polysynth ever was. Turn it up and each voice gets its own oscillator tuning error, its own cutoff offset, its own envelope times, a slow wandering drift and a fast jitter — separately per rank. A poly patch stops being one synthesizer played eight times and becomes eight slightly different synthesizers.

Most factory patches sit between 40 and 55 per cent. Very high is a machine that needs servicing, which is occasionally what you want.

MASTER TUNE · FINE · BEND RANGE

±12 SEMITONES · ±50 CENTS · 0 TO 24

These four, with **VOLUME** and **QUALITY**, are the only controls a factory patch does not touch — load a patch and your tuning, your bend range and your output level stay where you put them.

QUALITY

1X / 2X / 4X. Everything up to and including the tape runs at this rate. **2X** is the default and what the patches were made on. Eight voices with every stage running measure **2.5 per cent** of one core at 1x, **4.6** at 2x and **10.4** at 4x. It is not automatable, and it is not saved into a patch.

DRIVE, ENSEMBLE, CHOIR



The chain row. Drive, ensemble, choir, tape and hall, in the order the sound meets them.

DRIVE

BEFORE THE DECIMATOR, SO IT CANNOT ALIAS

OFF	Where a fresh instrument sits.
VALVE	Asymmetric, biased, and therefore even harmonics. Warmth rather than dirt.
TAPE	A soft knee, symmetric. Thickens without changing the shape much.
FUZZ	Hard, with a tilt ahead of it. For leads that have to cut.
FOLD	A sine folder. Past a point it stops being distortion and becomes a different instrument.

DRIVE TONE is a tilt afterwards, bipolar. Each mode carries a fixed trim measured on the bench, so turning DRIVE up changes what it sounds like rather than simply making it louder.

ENSEMBLE

THREE TAPS PER CHANNEL, TWO LFOS, 120 DEGREES APART

CHORUS I	Two taps, the slow LFO only. Gentle.
CHORUS II	Two taps, the fast wobble. Closer to a flutter than a chorus.
ENSEMBLE	Three taps, both LFOS. The string machine.
CATHEDRAL	Three taps, both LFOS, deeper, the wet path darkened as a bucket brigade darkens it.

CHOIR

FIVE FORMANTS, $U \rightarrow O \rightarrow A \rightarrow E \rightarrow I$

A fixed bank of five band-passes at the formants of a vowel. **VOWEL** slides continuously through all five — it is a position, not a selector, and everything between two vowels exists. **REGISTER** shifts the whole bank up to 40 per cent either way, which is the difference between a man and a boy. **AIR** adds breath on top. **MIX** is how much of it you want.

Feed it a pulse-rich rank and it is a human voice. Put the ensemble behind it and it is a choir.

MEASURED

Three modulated taps partly cancel — **-4.3 dB** at full mix — so the wet path is normalised between the amplitude sum and the power sum rather than by either.

THE TAPE, AND THE HALL

TAPE — THREE POSITIONS

OFF	Straight through.
TAPE	Wow, flutter, a saturator with a head bump, loss with age, dropouts and hiss.
VHS	Head-switching instead of flutter: a 50 Hz tracking wobble, a tick per field, bandwidth capped near 6 kHz, dropouts far more often.
Wow · Wow Rate	a 0.1 to 3 Hz sine plus a slewed walk — a wander, never a step
Flutter	sines at 4.3, 8.7 and 13.1 Hz, plus a fast random part
Tape Sat	the saturator and its head bump
Tape Age	how much top the tape has lost
Dropouts	raised-cosine dips with extra loss
Hiss	added last, never inside a loop

MEASURED

WOW at full and 0.5 Hz moves the pitch **2.15 per cent** peak to peak; the sine alone would give 1.1 and the walk adds the rest. **FLUTTER** at full: **0.35**. Both at zero: **0.0003** — nothing. **HISS** and **AGE** at full: **−57 dBFS** in the second after a note, and exactly silent when nothing plays.

A 2 s **DECAY** measures **1.68 s**. At maximum decay, size, shimmer and motion the tail is **97 dB** lower forty seconds after the note than in its first seconds: bounded, and falling.



The tape window. The reels turn while the tape runs.

HALL

EIGHT LINES BEHIND FOUR DIFFUSERS

Mix · Predelay	wet share; 1 to 250 ms
Size	half to double
Decay	a real RT60, 0.2 to 20 s
Damping	1 to 16 kHz, per line
Motion	two lines never stand still
Shimmer	an octave up, budgeted against the decay

BRASS, STRINGS, CHOIR, LEAD, BASS

BRASS

Blade Brass	Saw, pulse and sine on both ranks, rank II nine cents sharp; ensemble, tape and a 2.4 s hall. Touch opens it.
Chariots Brass	Full saws, rank II a 16' ladder underneath six cents flat; a wide filter swing into a 3.2 s hall.
Tight Brass	Two 8' saws through closed ladders, 70 ms attack, short decay. Velocity straight into the brightness.
French Horns	Saw, pulse and sine, rank II at 16', both filters dark; a 300 ms swell and a 4.5 s hall.
Brass Section	DUO mode with eight cents of detune: two players per key, wide, in a 3 s hall.

STRINGS

CS Strings	Two saws, rank II at 16' and eight cents sharp, high-passed at 120 Hz; ensemble at 55 per cent, a 3.5 s hall.
VP Strings	Saw and pulse, high-passed at 200 Hz, filters wide open. Ensemble at 75 per cent depth and 70 wet: the string machine on its own.
Cathedral Strings	A 900 ms swell and a 2.5 s release, rank II at 4'; CATHEDRAL ensemble and a 7 s hall with shimmer.
Solina Room	Bright saws high-passed at 300 Hz, ensemble at 80 per cent wet, a small 1.2 s room, and a tape that drops out.

CHOIR

Human Voice	Pulse-led ranks through the formant bank at 85 per cent, vowel just past the middle. One voice, not yet a choir.
Vangelis Choir	Choir at 90 per cent on a dark vowel with the register lowered; CATHEDRAL ensemble and an 8 s hall with shimmer.
Boys Choir	Both ranks at 4', choir at 90 per cent on a close vowel, register up, plenty of air; a 5 s hall.

LEAD

Blade Lead	MONO, legato, 90 ms glide. Saw with sine, rank II eleven cents sharp; vibrato on the wheel and on touch.
Sync Lead	MONO with glide. Rank I is synced to a silent rank II and starts seven semitones up; the filter envelope sweeps it. A little VALVE.
Screaming Lead	UNISON at fourteen cents, both ranks through resonant ladders at 900 Hz, FUZZ at 45 per cent.
Duo Lead	Two voices per key twelve cents apart, 70 ms glide, ladders at 1.8 kHz, vibrato on touch.

BASS

Vintage Bass	MONO legato. Saw with pulse and a lot of sine, ladders at 500 and 400 Hz, 8 ms attack, TAPE drive.
Moogish Bass	MONO legato, two saws, ladders at 350 Hz with the resonance up, 5 ms attack and a fast decay.
Pulse Bass	MONO legato. A 70 per cent pulse with PWM from a 0.9 Hz sub-oscillator, a little VALVE, almost no hall.

PAD, KEYS, BELLS, STRANGE

PAD		BELLS	
Heaven and Hell	Saw, pulse and sine with PWM from a 0.25 Hz sub-oscillator, a 1.8 s filter attack; CATHEDRAL, a 9 s hall with shimmer.	Ring Bells	Two triangle ranks a fourth apart at 4' and 2', RANKS ring modulation at 80 per cent, no sustain; a 6 s hall with shimmer.
Antarctic Pad	Saw with triangle and a little noise, 2.5 s attack, high-passed at 200 Hz; a 0.12 Hz triangle on the cutoff and a 14 s hall.	Tubular Dream	Triangle and pulse with a key-tracked carrier sweeping downward over a 900 ms ring decay; an 8 s hall.
Glass Pad	Triangle and sine at 4' and 2', key-tracked ring modulation at 35 per cent; a 7 s hall with shimmer at 40.	Sequencer Pluck	Two saws, ladders at 500 Hz with the envelope full open, 1 ms attack, no sustain, 60 ms of predelay.
Tape Wash	Saws with a little noise and a 3 s attack, onto a tape at 70 per cent wow, 75 age, dropouts and hiss; a 12 s hall.	STRANGE	
Wide Unison Pad	UNISON at twenty-two cents with the spread at full; slow filters and a 6 s hall.	Poly-Mod Sweep	Rank II is a silent triangle modulator: 45 per cent into rank I's pitch, 30 into both filters, and the envelope on top.
The Planets	16' and 8' saws with sine, filters closed to 600 Hz, a 0.15 Hz sub-oscillator on the cutoff; a 12 s hall.	Stranger Things	Two 16' saws twelve cents apart through resonant ladders, a short hard envelope, VALVE drive and the VHS.
KEYS		VHS Memory	Saw, pulse and sine on a worn VHS: 60 per cent wow, 60 flutter, 25 dropouts, 55 hiss, CATHEDRAL, a 6 s hall.
Electric Grand	A struck rank I with a 2 ms attack and a long decay, a bright 4' triangle rank II with no sustain; both hard on velocity.	Insect Swarm	UNISON at forty cents, 94 per cent pulses with noise, high-passed at 800 Hz; a noise sub-oscillator at 12 Hz and a 37 Hz ring.
Clavinet 84	Narrow pulses at 88 and 92 per cent, high-passed at 200 Hz, resonant ladders, 1 ms attack, no sustain, VALVE.	Machine Hum	MONO. A 16' rank closed to 180 Hz with a 3 s attack, a 0.08 Hz triangle on the cutoff, TAPE drive and a 9 s hall.
Organ 84	Sine and pulse at 8' and 4', filters open, no envelope to speak of, and a 6.5 Hz tremolo shared by every voice.	Filter Sweep Drone	Saw and pulse with PWM through resonant ladders at 400 Hz, a 4 s filter attack and a 10 s hall.
Soft Keys	Saw, triangle and sine, a 4 ms attack and a long decay to a low sustain; CHORUS I and a 2.8 s hall.	Ghost Hall	Triangle, sine and noise at 4' and 8', the choir half in on a dark vowel, VHS at 70 per cent age, a 16 s hall with shimmer.
		Naked Rank I	One 8' saw, filters open, a gate instead of an envelope, no hall and no vintage. The engine with nothing on it.

PATCHES ON DISK



The display. Amber, and the only place in the instrument that prints a number: the patch, the two filter envelopes, the eight voice lamps and the output.

The forty on the dial are read-only. Everything you make goes to disk as plain JSON, in the house folder every Brokild instrument uses:

DOCUMENTS \ BROKILD PATCHES \ 1984

SAVE and **OPEN** are ordinary file dialogs, so you can keep patches in folders or send one to somebody; the browser beside them lists the house folder, subfolders and all, without a dialog. A patch file carries the whole instrument: all 126 parameters, **and the Brokild World FX rack blob inside it**, so a patch that was made with an echo and a room in the rack comes back with them. The performance controls do not travel — master tune, fine, bend range, volume and quality stay as you have them when a patch loads.

RANDOM

RANDOM rolls a whole instrument, not a spray of values: the oscillators land on shapes that go together, the filters inside musical ranges, the envelopes in usable times, and the master and the performance controls are left alone.

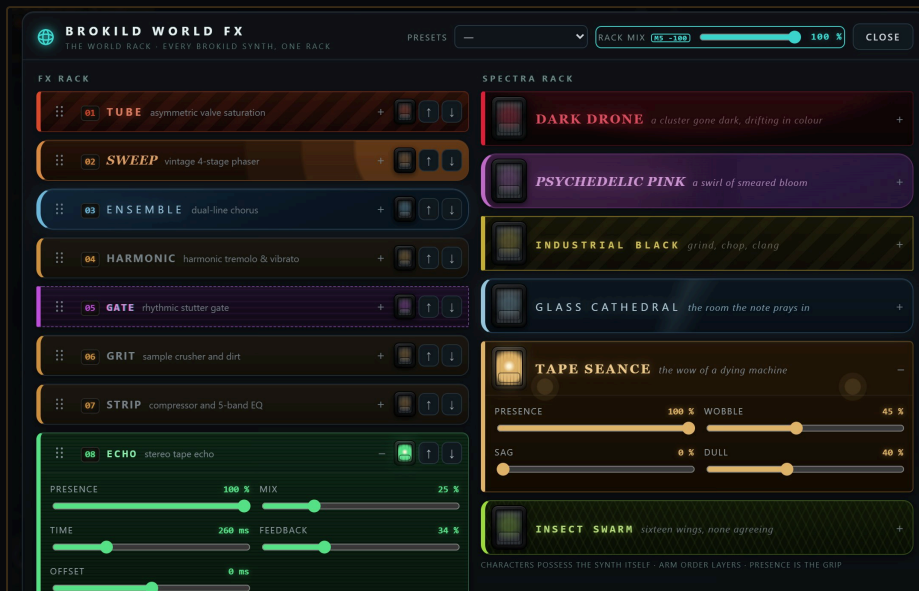
WHAT THE NUMBERS MEAN

Frequencies and times run on an exponential sweep, so a slider is fine where it needs to be: a cutoff covers 20 Hz to 20 kHz, an attack 1 ms to 10 s. A bipolar control reads -100 to +100 per cent about a centre of zero, tuning is in cents, width 50 to 95 per cent, and a glide under one per cent reads **OFF**.



The patch strip. The dial, the patch name and its category, with the browser and RANDOM, SAVE, OPEN, USER and PANIC beside it. Move any control and its value replaces the name, then the name comes back.

BROKILD WORLD FX, AND THE FIVE MACROS



The rack. Effects down the left in the order they run, SPECTRA characters down the right, and the rack's own mix in the header. ECHO is open; every pedal unfolds like it.

TWO ROOMS IN A ROW

1984 has a hall of its own and the rack has a SPACE. Running both is legitimate and occasionally wonderful, but the usual answer is to pick one: the instrument's hall for the sound of the machine, the rack's for the sound of the record.

Every Brokild instrument carries the same rack at the end of its chain, and 1984 is no exception. Open it with the globe and you get a pedalboard: a column of effect modules you can switch on, reorder and set, and a second column of SPECTRA characters that reach into the instrument itself rather than filtering its output. A fresh instrument has an empty rack and sounds exactly as though the rack were not there; a patch you save carries whatever you built in it. It is the same rack, with the same modules and the same presets, as the one in every other machine on the shelf, so a chain you like moves with you.

BWFX MACRO 1 to 5 are the rack's only host parameters, and they are the reason it can be automated at all. A macro **maps** a control in the rack: assign one and it takes that control from the bottom of its range to the top as the macro sweeps from 0 to 100 per cent, or the other way round with a negative depth, and the control visibly moves as it does. One destination belongs to one macro. **Macro 5 arrives assigned to the rack's own dry/wet at full negative depth**, so a fresh instrument with the macro at rest is exactly the sound it always was, and pushing it up is one global effects fader.

WHERE THE RACK SITS

After the hall and before the output. The instrument's own drive, ensemble, choir, tape and hall are not part of it and do not duplicate it — they are the machine, and the rack is what you put the machine through.

SPECIFICATIONS

Format	Windows VST3 (64-bit), plus a standalone application
Type	Instrument — MIDI in, stereo out
Voices	Eight, each of two complete ranks — sixteen ranks in all
Oscillator	Per rank: band-limited saw, pulse, triangle and sine at once, each with its own level, plus white noise. 32' / 16' / 8' / 4' / 2', ±12 semitones, ±50 cents
Pulse width	50 to 95 per cent, with PWM from the sub-oscillator and from poly-mod
Filters	Per rank: resonant 12 dB high-pass, 10 Hz to 12 kHz, into a resonant low-pass, 20 Hz to 20 kHz, switchable between a 12 dB state-variable circuit and a 24 dB ladder. One filter envelope drives both
Envelopes	Per rank: a filter envelope with initial level, attack level, attack, decay and release (1 ms to 10 s, 2 ms to 20 s), and an ADSR amplifier envelope
Sync	Rank I hard-synced to rank II, band-limited at the actual jump height
Poly-mod	Rank II pre-filter into rank I pitch, width and both cutoffs; rank I's filter envelope into its own pitch and width. Per voice, at audio rate
Ring modulator	Per voice. A carrier of its own from 0.1 Hz to 4 kHz, or a ratio of the note from 0.25x to 8x, with attack, decay and a bipolar sweep; or rank I multiplied by rank II

Sub-oscillator	Per voice. Seven shapes, 0.05 to 40 Hz, delay to 5 s, phase free, reset or shared, into pitch, width, filter and level
Touch	Velocity into level and brilliance; aftertouch, channel or polyphonic, into pitch, brilliance, level and the sub-oscillator; mod wheel into vibrato and brilliance at its own 2 to 10 Hz
Voice modes	Poly 8, Duo, Unison 8 with up to 60 cents of detune, Mono with legato. Glide 5 ms to 5 s, optionally glissando
Chain	Drive (valve, tape, fuzz, fold), ensemble (four modes), choir (five formants, U to I), tape or VHS, hall (eight-line FDN, 0.2 to 20 s, predelay to 250 ms, shimmer)
Oversampling	1x, 2x or 4x up to and including the tape; the hall runs at the base rate. 2x by default
Parameters	126, plus the five BWFX macros. All automatable except the patch dial and the quality switch
Patches	Forty factory, in nine categories. Your own are JSON in DOCUMENTS \ BROKILD PATCHES \ 1984 , carrying the rack with them
CPU	Eight voices, every stage running: 2.5 per cent of one core at 1x, 4.6 at 2x, 10.4 at 4x
Sample rates	Any. Bench-tested at 44.1, 48 and 96 kHz
Price	Free. No account, no registration, no telemetry

WHAT THE BENCH PROVES

Eighty-six checks, all clear, at 48 kHz and again at 44.1 and 96. The bench renders real audio and measures it. These are its numbers, and they are the only numbers in this handbook.

TUNING	Both ranks within one cent at every footage. Rank II asked for +25 cents lands on +25.0 .
ALIASING	A saw at A6, worst non-harmonic component below 20 kHz: under -40 dB at 1x and under -60 dB at 2x. A 5 per cent pulse at 2x, under -50 dB . FUZZ at full drive at 2x, under -36 dB .
FILTERS	Asked to self-oscillate at 1 kHz, the state-variable circuit sits at 1000.0 Hz and the ladder at 996.5 Hz — 0.35 per cent — and both stay bounded. Resonance at 80 per cent: 14.5 dB over the octave above.
VALVE DRIVE	A second harmonic of -16 dB at 30 per cent drive. An asymmetric clipper with unequal tops is DC and odd harmonics; the even harmonics need an even term, and this one has it.
ENSEMBLE	Three modulated taps partly cancel — -4.3 dB at full mix — so the wet path is normalised between the amplitude sum and the power sum.

TAPE	WOW at full and 0.5 Hz: 2.15 per cent peak to peak. FLUTTER at full: 0.35 per cent . Both at zero: 0.0003 per cent . HISS and AGE both at full: -57 dBFS in the second after a note, and exactly silent when nothing is playing.
HALL	A 2 s setting measures 1.68 s . At maximum decay, size, shimmer and motion the tail is 97 dB lower forty seconds after the note than in its first seconds — bounded, and still falling. Stopping the input and watching is the only test that tells a tail from an oscillator.
COST	Eight voices with every stage running: 2.5 per cent of one core at 1x, 4.6 at 2x, 10.4 at 4x.
AND	Silence at rest; the envelope shapes; the velocity range; sync making harmonics at the master; ring making sums and differences; poly-mod moving the spectrum; unison beating at the detune; glide times; eight notes sounding and a ninth stealing the oldest; mono legato; choir formant peaks; dropouts counted; DC; two hundred random machines bounded; renders identical to the byte twice over; every parameter driven to both extremes.

PROBES THAT LIED BEFORE THE CODE DID

Six of them, and each was rewritten rather than loosened: a zero-crossing pitch estimator that read 46 per cent of wobble out of a two-tap chorus's own cancellation nulls; a pulse source with a spectral null sitting exactly on the formant under test; a pluck measured after it had already decayed; a glide test reading voice 0 of a polyphonic allocation; a vintage test on the one voice whose tolerance happened to be small; and a sustain test reading past the end of its own buffer.

1984

The CS-80, the Prophet-5 and the VP-330 are the machines it is after, and it copies none of them.

What it takes from each is the idea, not the circuit. From the first: two ranks per voice, a high-pass in series with a low-pass, and a filter envelope built out of an initial level and an attack level instead of a sustain. From the second: hard sync, poly-mod and unison, and the habit of letting an oscillator modulate a filter at audio rate. From the third: a fixed formant bank behind a three-phase ensemble, which is how a machine can sound like people.

Everything in it was written from scratch, measured on a bench, and kept only if the measurement agreed with the claim. Nothing was tuned by ear afterwards and then described as physics.

WHERE TO START

CS Strings for what the architecture is for. **Vangelis Choir** for the choir and the hall. **Poly-Mod Sweep** for the strange half. **Naked Rank I** for one oscillator with nothing on it at all, which is where to go when you want to build something yourself.

Instrument	1984, a Brokild polysynth
Build	260923.1
Written by	Peter Bøggild, 2026
Built with	JUCE 8, native C++ audio, the interface drawn in the plugin's own window
World rack	Brokild World FX, the same rack every Brokild instrument carries
Home	peterboggild.github.io/BrokildApps

THE USUAL WARNING

This VST comes from the curved mind of Professor Brokild, and was programmed with VScode, ChatGPT and Claude. Use of this is on your own peril. It may generate unique, beautiful, scary or useless sounds for you, and it may crash your DAW project. Have fun, play unsafely, unwisely, and unboringly.

A LATE NIGHT IN 1984